econ0004 - test

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Jan 31, 2026

Multiple Choice Questions (15 questions)

Choose the best answer for each question.

1. Which of the following is an example of **time series** data?

   (a) Data on the consumption of 200 households during 2023.

   B (b) Data on the GDP of a country from 2000 to 2020.

   (c) Data on the birth rate and death rate in 5 countries over 10 years.

   (d) Data on the test scores of 100 students in 2024.

2. A data set that follows the same group of individuals (e.g., households, firms) over multiple time periods is called:

   (a) Cross-sectional data.

   B (b) Time series data.

   (c) Panel data.

   (d) Pooled cross-sectional data.

3. Pooled cross-sectional data differ from panel data in that:

   (a) Pooled data follow the same units over time, while panel data use different units each period.

   B (b) Panel data follow the same units over time, while pooled data use different units each period.

   (c) Pooled data contain only one variable, while panel data contain many variables.

   (d) There is no difference; the terms are interchangeable.

4. Econometricians typically work with:

   (a) Experimental data from controlled laboratory settings.

   B (b) Nonexperimental (observational) data.

   (c) Only time series data.

(d) Only cross-sectional data.

5. Which of the following is a typical goal of econometric analysis?

   (a) To prove that correlation implies causation.

   B (b) To estimate the causal effect of one variable on another using observational data.

   (c) To avoid using statistical methods.

   (d) To collect only experimental data.

6. The statement "Correlation does not imply causation" means that:

   (a) If two variables are correlated, one must cause the other.

   C (b) If two variables are correlated, there is no possibility of a causal relationship.

   (c) A statistical association between two variables does not necessarily mean one causes the other.

   (d) Causation can only be established by calculating correlation coefficients.

7. Which scenario best illustrates **reverse causality**?

   (a) Higher ice cream sales are associated with more drownings because hot weather increases both.

   B (b) Cities with more police officers have higher crime rates because crime causes more police to be hired.

   (c) More educated people read broadsheet newspapers and have more tolerant views on immigration.

   (d) An increase in the minimum wage leads to higher unemployment.

8. An **omitted variable** is a variable that:

   (a) Is included in the regression but has no effect on the outcome.

   B (b) Is not observed but influences both the treatment and the outcome, creating a spurious correlation.

   (c) Is the main cause of the outcome and is always measured perfectly.

   (d) Causes reverse causality.

9. In the newspaper readership example (UK data), the initial finding was that:

   (a) Tabloid readers are less hostile to immigration than broadsheet readers.

   C (b) Broadsheet readers are more hostile to immigration than tabloid readers.

   (c) Tabloid readers are more hostile to immigration than broadsheet readers.

   (d) There is no relationship between newspaper type and immigration attitudes.

10. After controlling for education (A-level qualification) in the newspaper example, the relationship between newspaper type and immigration attitudes:

B

(a) Completely disappeared, suggesting education was the only driver.

(b) Persisted within education groups, suggesting that newspaper readership may have an independent effect or other confounders exist.

(c) Reversed: tabloid readers became less hostile.

(d) Became stronger only for the less educated group.

11. Based on the overall admission rates by race at private institutions (from the slides), which group had the lowest admission rate?

D

(a) White

(b) Black

(c) Hispanic

(d) Asian

12. When admission rates were compared within the same SAT score intervals, the data showed that:

(a) Asian applicants had the highest admission rate in every score interval.

B

(b) Asian applicants had the lowest admission rate in every score interval.

(c) There were no differences across races within the same score interval.

(d) White applicants had the lowest admission rate in the top score interval.

13. The term "ceteris paribus" (other things being equal) is crucial in causal analysis because:

(a) item It allows us to ignore all other factors and focus only on the variable of interest.

B

(b) It reminds us that to isolate a causal effect, we must hold other relevant factors constant.

(c) It is the same as the concept of correlation.

(d) It applies only to experimental data.

14. Which of the following is an example of using econometrics for **policy evaluation**?

(a) Estimating the elasticity of demand for a product.

D

(b) Testing the law of demand using historical data.

(c) Forecasting next year's GDP growth.

(d) Assessing the impact of a minimum wage increase on employment.

15. Why is it difficult to infer causality from nonexperimental data?

(a) Because we cannot observe all relevant variables and cannot randomly assign treatments.

A

(b) Because economic data are always measured with error.

(c) Because correlation coefficients are always close to zero.

(d) Because we cannot use statistical methods.

True/False Questions (5 questions)

Indicate whether each statement is True or False.

F 1. If two variables are correlated, then there must be a causal relationship between them.

T 2. Panel data consist of observations on the same cross-sectional units over a given period of time.

F 3. In the newspaper example, after controlling for education, the relationship between newspaper type and immigration attitudes disappeared, proving that education was the only confounder.

F 4. Overall admission rates by race are sufficient evidence to prove discrimination in college admissions.

T 5. One of the goals of econometrics is to forecast economic variables such as future GDP growth.

Calculation Questions (2 questions)

Show your work.

Question 1: Newspaper Readership and Immigration Attitudes

The following table summarizes data from the UK newspaper readership study (hypothetical numbers based on the lecture).

Full Sample			
Preference	Tabloid	Broadsheet	All
Same or more	931	297	1228
Less	2199	373	2572
Total	3130	670	3800

By Education Group						
	A-level holders			Non A-level holders		
	Tabloid	Broadsheet	All	Tabloid	Broadsheet	All
Same or more	106	136	242	397	49	446
Less	221	115	336	880	70	950
Total	327	251	578	1277	119	1396

1. From the full sample, compute the percentage of tabloid readers who prefer less immigration. Also compute the same percentage for broadsheet readers. $\frac{2199}{2572} \times 100\% = 85.5\%$ $\frac{373}{2572} \times 100\% = 14.5\%$

2. What is the difference in these percentages (tabloid minus broadsheet)?
$85.5\% - 14.5\% = 71\%$

3. Among A-level holders, compute the percentage of tabloid readers who prefer less immigration, and the same percentage for broadsheet readers. What is the difference? $\frac{221}{336} \times 100\% = 21.5\%$ $\frac{115}{336} \times 100\% = 34.2\%$ $34.2\% - 21.5\% = 12.7\%$

4. Based on your answers in (c), does controlling for education eliminate the association? What can you conclude?
No. because the percentage differences, 14.5% and 12.7% are very close.

Question 2: College Admissions by Race and SAT Scores

Consider the following hypothetical admission rates for private universities, broken down by race and SAT score intervals.

SAT Score Interval	White	Black	Asian
1400–1600	30%	45%	25%
1200–1399	20%	28%	18%
1000–1199	10%	15%	8%

1. In the highest SAT interval (1400–1600), what is the admission rate for Asian applicants? For White applicants?
Asian: 25% White: 30%

2. Calculate the difference in admission rates between White and Asian applicants in the top interval (White minus Asian). $30\% - 25\% = 5\%$

3. Explain why comparing admission rates within the same SAT interval is more informative than comparing overall admission rates when investigating possible discrimination.
By controlling a confounder (SAT interval 1400~1600), the correlation may be more reliable.

Essay Question (1 question)

Write a concise answer (about 200–300 words).

Discuss the newspaper readership example. Why can we not conclude that reading tabloid newspapers *causes* more hostile attitudes toward immigration based solely on the observed correlation? Identify at least two possible confounding factors or alternative explanations, and suggest how one might use data to further investigate causality.

It is clear to see that the proportion of people who preper reducing immigrations reading tabloid (70.3%) is greater than that reading broadsheets (55.7%), but we cannot conclude that there's a causal relationship between reading tabloids and anti-immigration attitude. The observed difference may be driven by omitted variables such as education, which may influence readers' choices toward the immigration. In addition, the reverse causality is possible as individuals may choose newspapers that align with their own existing beliefs. To further investigate causality, researchers could control some variables and examine the changes over time with panel data, in order to isolate the external effects such as medias.